

Alexander Winkler

awinkler.dev | Alexwinkler1@icloud.com | [linkedin.com/in/alexwinklerr](https://www.linkedin.com/in/alexwinklerr)

Education

Georgia Institute of Technology

Master of Science in Computer Science

Jan. 2026 – Dec. 2027

Online

Eastern Florida State College

Bachelor of Applied Science in Software Development (Deg GPA: 4.0)

May 2021 – Aug. 2025

Melbourne, FL

Experience

Cloud Architecture Intern

Georgia Technology Authority

Jun. 2026 – Aug. 2026

Atlanta, GA

- Incoming Cloud Architecture Intern tasked with defining deployment paradigms across **Docker**, **AWS**, and **GCP** to optimize enterprise-scale distributed topologies and multi-tenant API boundaries.
- Architect decoupled microservices frameworks and high-throughput backend infrastructure, establishing core patterns for high availability and low-latency fault tolerance.

Teaching Assistant – Android App Development

Eastern Florida State College

Jan. 2025 – May 2025

Melbourne, FL

- Mentored students through **8+** complex Android projects, providing technical code reviews and debugging support in **Kotlin** and **Java**.

Police Officer

City of Rockledge

Jan. 2022 – Jan. 2023

Rockledge, FL

- Operated in high-pressure field situations requiring clear communication, situational awareness, and accurate report writing

Projects

Amazon Enterprise Security Analytics Engine | *AWS RDS, MySQL, FastAPI, Next.js, Python, SQLite* [Link](#)

- Architected a search platform to analyze a **1.2M+ record breach dataset**, uncovering data exposure risks affecting executive personnel and reporting the findings to AWS Security, leading to remediation
- Engineered a migration from local storage to **Amazon RDS (MySQL)**, optimizing query latency and building a scalable infrastructure for real-time risk assessment.
- Leveraged **serverless Python** backends to handle high-concurrency traffic with minimal maintenance overhead and sub-second retrieval times.

Ranking Algorithm Integrity Framework (Zillow Audit) | *Python, Next.js*

[Link](#)

- Reverse-engineered engagement-based ranking systems to identify logic flaws in automated verification and rate-limiting controls.
- Built a high-concurrency simulation suite that increased property visibility by **22,000% (2 to 450+ views)** and generated **400 synthetic interactions** to audit "Hot Home" triggers.
- Published a **10-page technical research paper** detailing system vulnerabilities and architecting defensive mitigations against engagement fraud.

CrisisCompass (UN × Databricks Challenge) | *Databricks, PySpark, React, FastAPI, Vite, Mapbox GL JS, XGBoost* [Link](#)

- Built a real-time humanitarian intelligence platform in **36 hours**, analyzing disaster risk vs. funding across **53 countries**
- Processed **1,400+ disaster records** using **PySpark** and **distributed ETL pipelines** to identify **underfunded high-risk crises**
- Implemented **AI-generated briefings** with risk scores, historical patterns, and actionable recommendations

Technical Skills

Languages: Python, Java, C++, SQL, JavaScript, Kotlin, HTML/CSS

Cloud & DevOps: AWS (RDS), Google Cloud Platform, Docker, Vercel, Git

Data & Backend: PySpark, Databricks, FastAPI, MySQL, SQLite

Frameworks & Tools: React, Next.js, Android Studio, BeautifulSoup, PlantUML, Bruno